



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105



Student Name: - SANDRA SUDEVAN

Reg. No: 192424422

Course Code: CSA0509

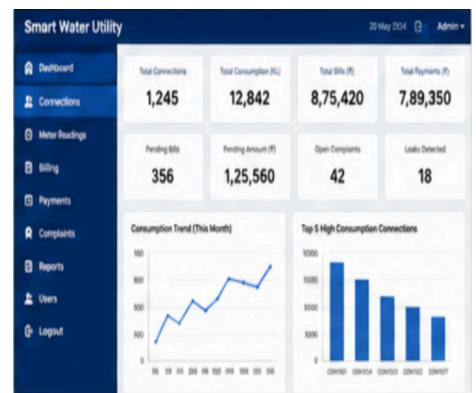
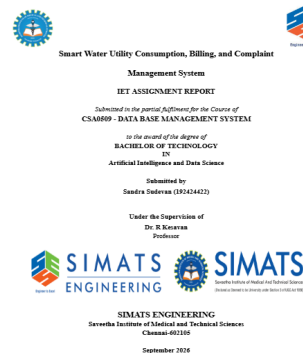
Slot: A

Course Name: DATABASE MANAGEMENT SYSTEMS

Course Faculty: Dr. R. Kesavan

Project Title: Smart Water Utility Consumption, Billing, and Complaint Management System

Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: (here you write what you did in this project (contribution) including Model Description)

The **Smart Water Utility Consumption, Billing, and Complaint Management System** is a web-based database application developed to make water utility operations faster, more reliable, and easier to manage. The system collects and stores smart-meter readings along with connection details, timestamps, and consumption values, allowing staff to quickly view consumption history and identify unusually high usage that may indicate water leakage. To improve data retrieval, the system uses B+ tree indexing and hash-based lookup techniques. It also manages monthly bill generation, payment recording, and customer complaints through a centralized database. Transactions, locking, and database constraints are used to prevent problems such as duplicate bills, incorrect payment updates, and lost complaint updates when multiple operators work at the same time. The system is implemented using MySQL with a user-friendly web dashboard that displays important information such as total connections, water consumption, pending bills, payments, complaints, and leak alerts. Overall, the project provides an efficient and scalable solution for managing large amounts of water utility data while supporting accurate billing, faster leak detection, and better service management.

Student Signature

Guide Signature